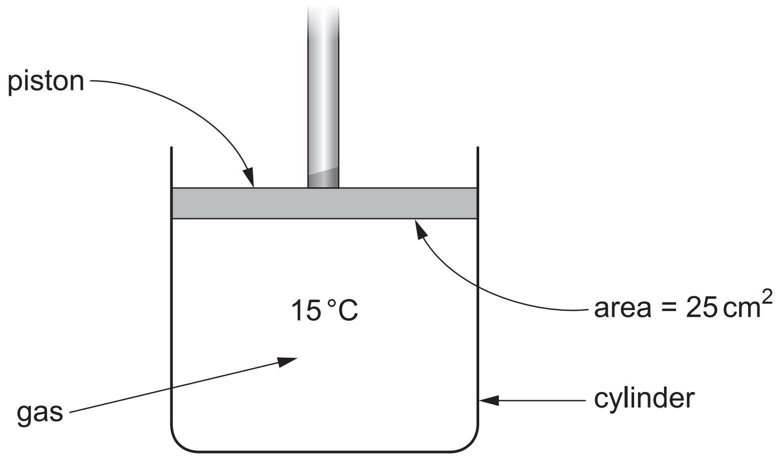
Examiner only

10. A piston holds $125\,cm^3$ of gas in a cylinder at a temperature of 15 °C.
The surface area of the lower surface of the piston is $25\,cm^2$.
The gas exerts a force of 250 N on this surface.



Use the information above to answer the following questions.

(a) Use an equation from page 2 to calculate the pressure of the gas inside the cylinder. [2]

pressure = ... N/cm^2

(b) The temperature of the gas increases to 298 K and the gas is compressed so the volume becomes $75\,cm^3$.

(i) Convert **15 °C** into K.
Use the equation: [1]

$$T/\text{K} = \theta/°\text{C} + 273$$

T = ... K

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(ii) Use the equation:

$$\frac{pV}{T} = \text{constant}$$

to calculate the new gas pressure. [4]

pressure = ... N/cm^2

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END OF PAPER